

MyTZStudies.com

Your Free Tanzania Exam Library

Past papers, answer keys, and study resources
for Tanzanian students.

MyTZStudies.com

Past Exam Papers	Worked Answer Keys	Free Access
Choose Your Level	Study Step by Step	Learn at Your Pace

Standards 1-7 | Form 2 | Form 4

Share this with a friend.
Every student deserves free study materials.

MATHEMATICS

STANDARD SIX

Monthly test - September 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Six
Year	2024
Exam Board / Issuer	SPES
Exam	Monthly test - September
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2024-SPES-September-Monthly-test.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Calculations

1(i). Add $489685 + 19867$.

Step 1: 1 place: $5 + 7 = 12$; write 2, carry 1.

Step 2: 10 place: $8 + 6 + \text{carry } 1 = 15$; write 5, carry 1.

Step 3: 100 place: $6 + 8 + \text{carry } 1 = 15$; write 5, carry 1.

Step 4: 1000 place: $9 + 9 + \text{carry } 1 = 19$; write 9, carry 1.

Step 5: 10000 place: $8 + 1 + \text{carry } 1 = 10$; write 0, carry 1.

Step 6: 100000 place: $4 + 0 + \text{carry } 1 = 5$; write 5.

Step 7: $489685 + 19867 = 509552$.

Final Answer: 509552

1(ii). Subtract $700505 - 386785$.

Step 1: 1 place: $5 - 5 = 0$.

Step 2: 10 place: regroup 1 from the next place; $10 - 8 = 2$.

Step 3: 100 place: regroup 1 from the next place; $14 - 7 = 7$.

Step 4: 1000 place: regroup 1 from the next place; $9 - 6 = 3$.

Step 5: 10000 place: regroup 1 from the next place; $9 - 8 = 1$.

Step 6: 100000 place: $6 - 3 = 3$.

Step 7: $700505 - 386785 = 313720$; check $313720 + 386785 = 700505$.

Final Answer: 313720

1(iii). Multiply 455×38 .

Step 1: Split into place values: $400 \times 38 = 15200$; $50 \times 38 = 1900$; $5 \times 38 = 190$.

Step 2: Add the partial products: 17290.

Final Answer: 17290

1(iv). Divide 6561 by 81.

Step 1: $656 \div 81$: write 8; $8 \times 81 = 648$; $656 - 648 = 8$.

Step 2: $81 \div 81$: write 1; $1 \times 81 = 81$; $81 - 81 = 0$.

Step 3: $81 \times 81 = 6561$.

Step 4: $6561 \div 81 = 81$.

Final Answer: 81

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(ix). Divide 5.5 by 0.05.

Step 1: Multiply both numbers by 100: $550/5$.

Step 2: $550 \div 5 = 110$.

Final Answer: 110

1(v). Add $1 \frac{1}{5} + 6 \frac{1}{4}$.

Step 1: $\frac{6}{5} = \frac{24}{20}$; $\frac{25}{4} = \frac{125}{20}$.

Step 2: $24 + 125 = 149$; $\frac{149}{20} = 7 \frac{9}{20}$.

Final Answer: $7 \frac{9}{20}$

1(vi). Subtract $8 \frac{1}{4} - 3 \frac{5}{6}$.

Step 1: $\frac{33}{4} = \frac{99}{12}$; $\frac{23}{6} = \frac{46}{12}$.

Step 2: $99 - 46 = 53$; $\frac{53}{12} = 4 \frac{5}{12}$.

Final Answer: $4 \frac{5}{12}$

1(vii). Subtract $15 - 3\frac{3}{5}$.

Step 1: $15 = 75/5$; $3\frac{3}{5} = 18/5$.

Step 2: $75 - 18 = 57$; $57/5 = 11\frac{2}{5}$.

Final Answer: $11\frac{2}{5}$

1(viii). Add $50.05 + 1.85$.

Step 1: Hundredths: $5 + 5 = 10$; write 0 and carry 1.

Step 2: Tenths: $0 + 8 + 1 = 9$; whole-number part: $50 + 1 = 51$.

Final Answer: 51.90

1(x). Multiply $-65 \times +35$.

Step 1: The product of numbers with opposite signs is negative.

Step 2: $65 \times 35 = 2275$.

Final Answer: -2275

2(i). Write nine hundred fifty-six million three hundred twenty-three thousand and three in numerals.

Step 1: $956000000 + 323000 + 3 = 956323003$.

Final Answer: 956323003

2(ii). Write 8050050 in words.

Step 1: $8050050 = 8000000 + 50000 + 50$.

Final Answer: Eight million fifty thousand and fifty

2(iii). Convert 0.85 to a percentage.

Step 1: $0.85 \times 100 = 85$.

Final Answer: 85%

2(iv). Write MMDXC in Arabic numerals.

Step 1: $MM = 2000$; $D = 500$; $XC = 90$.

Step 2: $2000 + 500 + 90 = 2590$.

Final Answer: 2590

2(v). Round 50.956 to 2 decimal places.

Step 1: The third decimal digit is 6, which is at least 5; raise the hundredths digit from 5 to 6.

Final Answer: 50.96

2(vi). Find the square root of 1024.

Step 1: $32 \times 32 = 1024$.

Final Answer: 32

3(i). Janeth walks 3.62 km and 0.241 km. Find the total distance.

Step 1: Align decimal places: $3.620 + 0.241$.

Step 2: Thousandths: $0 + 1 = 1$; hundredths: $2 + 4 = 6$; tenths: $6 + 2 = 8$; whole-number part: $3 + 0 = 3$.

Final Answer: 3.861 km

3(ii). Find the HCF of 14, 20 and 32.

Step 1: $14 = 2 \times 7$; $20 = 2^2 \times 5$; $32 = 2^5$.

Step 2: The only shared prime factor is 2.

Final Answer: 2

3(iii). Which is largest: 0.25, 12%, 2/5 or -1000?

Step 1: Convert to decimals: 0.25, 0.12, 0.4, -1000.

Step 2: 0.4 is the greatest.

Final Answer: 2/5

4(i). Convert 2500 ml to litres in decimal form.

Step 1: $2500 / 1000 = 2.5$.

Final Answer: 2.5 L

4(ii). Given that Mwinyi was born in 1926 and died in 2024, write the difference between the years in Roman numerals.

Step 1: $2024 - 1926 = 98$.

Step 2: $90 = XC$; $8 = VIII$.

Final Answer: XCVIII (98 years from the printed years)

4(iii). List all even numbers strictly between 18 and 26.

Step 1: 20, 22, 24.

Step 2: Include 18 and 26 only if inclusive endpoints are intended.

Final Answer: 20, 22, 24

5(i). Solve $4p - 7 = 2p - 11$.

Step 1: Subtract $2p$ from both sides: $2p - 7 = -11$.

Step 2: Add 7 to both sides: $2p = -4$.

Step 3: $p = -2$; check: $-8 - 7 = -4 - 11 = -15$.

Final Answer: $p = -2$

5(ii). Find the mean attendance for 105, 125, 145, 140 and 135 pupils.

Step 1: $105 + 125 + 145 + 140 + 135 = 650$.

Step 2: $650 / 5 = 130$.

Final Answer: 130 pupils

5(iii). Form a number with 6 tens, 1 hundredth and 0 thousandths.

Step 1: Other unspecified places are taken as 0.

Step 2: $60 + 0.01 + 0 = 60.01$.

Final Answer: 60.010 (the same as 60.01, assuming other places are 0).

6(i). List the positive common multiples of 4 and 6 below 30.

Step 1: The LCM of 4 and 6 is 12.

Step 2: The positive multiples 12 and 24 are below 30; 36 is not.

Final Answer: 12, 24 (also 0 if a nonpositive starting multiple is permitted).

6(ii). My brother is 3 years older than me and our ages total 37 years. Find my age.

Step 1: Let my age be a , so my brother's age is $a + 3$.

Step 2: $2a + 3 = 37$, so $2a = 34$.

Step 3: $34 / 2 = 17$.

Final Answer: 17 years

6(iii). Which number is both even and prime?

Step 1: 2 has exactly the factors 1 and 2.

Final Answer: 2

7(i). The masses of Tulinagwe, Doreen, Kelvin, Derrick and Bakari are 14, 15, 17, 15 and 19 kg respectively. Find their mean mass.

Step 1: $14 + 15 + 17 + 15 + 19 = 80$.

Step 2: $80 / 5 = 16$.

Final Answer: 16 kg

7(ii). Zakayo shares Sh 48000 using a pie chart: Timoth has 110° , Tausi 100° and Onesmo the remaining sector. Find Onesmo's amount.

Step 1: $360 - 110 - 100 = 150^\circ$.

Step 2: $48000 \times 150 / 360 = 20000$.

Final Answer: Sh 20000

8(i). A parallelogram has points A(-3, 1), B(-1, -2), C(4, 1) and D(6, M). Find M.

Step 1: If ABCD must be the cyclic order, diagonal AC has midpoint $x = (-3 + 4)/2 = 0.5$, while BD has midpoint $x = (-1 + 6)/2 = 2.5$. This is impossible.

Step 2: As an unordered set, the order A, B, D, C works: A and C share $y = 1$, while B and D share $y = -2$.

Step 3: Then $M = -2$, and the AB vector (2, -3) equals the CD vector (2, -3).

Final Answer: $M = -2$ if the vertices are unordered (order ABDC); there is no solution for cyclic order ABCD.

8(ii). A cylinder has base area 38.5 m^2 and height 1000 m. Find its capacity in litres.

Step 1: Volume = base area $\times$ height.

Step 2: $38.5 \times 1000 = 38500 \text{ m}^3$.

Step 3: $1 \text{ m}^3 = 1000 \text{ L}$; $38500 \times 1000 = 38500000$.

Final Answer: 38500000 L

8(iii). A square circumscribes a circle of radius 7 cm. Find the shaded area outside the circle, using $\pi = 22/7$.

Step 1: The square side equals the circle diameter, 14 cm.

Step 2: $14 \times 14 = 196 \text{ cm}^2$.

Step 3: Circle area = $(22/7) \times 7 \times 7 = 154 \text{ cm}^2$.

Step 4: $196 - 154 = 42$.

Final Answer: 42 cm^2

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Source Checks

Darasa Huru - SPES Standard Six (VI) Monthly Exams September 2024